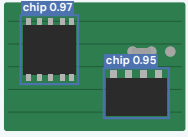
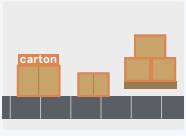
L-1: Closed-set detection (Sec. 3)



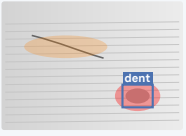
(1) Closed-set detection

Fixed label set $\mathcal{C}$ (Eq. (1)); boxes, classes and scores from a two-stage, YOLO or DETR detector; evaluated with COCO AP.



(2) Dense, real-time detection

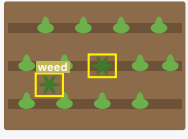
Many overlapping instances (SKU-110K: ∼147 per image); NMS or NMS-free one-to-one heads; latency budget per frame.



(3) Anomaly map → boxes

Unsupervised or zero-shot anomaly scoring (PatchCore, WinCLIP); thresholded regions become class-agnostic boxes (Sec. 7.1).

L-2: Open-vocabulary and grounded detection (Sec. 4)

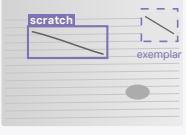


(1) Text-prompted OVD

Prompt: *"weed . crop"*. Region–text similarity (Eq. (2)) detects categories unseen in training; zero-shot LVIS AP / AP_r.

(2) Referring expression

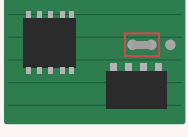
Q: *"the carton on top of the pallet"* → one box. Phrase grounding and described-object detection extend this to all matches and to negation.



(3) Visual-prompt detection

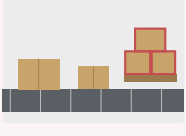
One annotated exemplar of a new defect replaces a text description (T-Rex2, YOLOE); suited to vocabularies that are hard to name.

L-3: MLLM-based and agentic detection (Sec. 5)



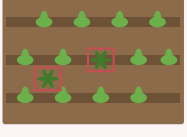
(1) Direct coordinate generation

Q: *"Locate the solder bridge."*
A:
`<box>(667,294,854,471)</box>`
Quantised tokens (Eq. (19)), scored by F1.



(2) Tool-orchestrated agent

Q: *"Is any carton on the pallet damaged?"* Plan → `detect(carton)` → crop → verify.
A: *"3 cartons, none damaged."*



(3) Compositional reasoning

Q: *"Which rows need spot spraying?"* Count weeds per row, apply a rule, return rows and nozzle commands with a logged trace.